Mandyoc outputs (NetCDF from tapioca)

lithology.nc, temperature.nc, strain.nc, surface.nc; interfaces.txt (rho0),
param.txt (α, sea level), SP_vc.txt / infos_*.txt (roles of the layers)

1. Load

roles of the layers (mantle, crust, basalt, sediments, water, air);
depths below sea level (top of the water layer or initial surface)

2. Fine grid

smoothed role indicators → largest wins (sharp contacts, no staircase);
age of each sediment layer; water above the seafloor (surface.nc)

3. Physical properties: density and Vp

crust, basalt, mantle: density rho0 (1 − α T); Vp per role − 0.4 m/s/K × T
sediments: compaction (Athy) + one impedance per deposited layer
+ serpentinization: mantle < 400 °C, < 4 km below the top of the basement
+ heterogeneity: von Kármán random media (laminated lower crust)
+ faults: localized strain → damage zone + fabric along the band (brittle)

4. Convolutional section

whole window (X0–X1)
depth → TWT (true velocities)
reflectivity (Z2 − Z1) / (Z2 + Z1)
∗ Ricker 25 Hz, + noise, top mute

5. Zoom model (XZ0–XZ1, DXZ)

fast impedance-matched water, receivers on the seafloor

_output/seismic_section_parallel_zoom_<window>.nc

exists, with the same parameters?

no

yes

Exploding reflector

Devito, 8th order (OpenMP)
half velocities, one simulation
→ zero-offset data + noise

Load .nc

skip the
simulations

Reverse-time migration

traces reversed in time,
smoothed model, Laplacian
→ depth image; save .nc

6. Display and outputs (_output folder of the scenario)

depth → TWT with true velocities; AGC, grey scale; figures (model, convolutional,
zoom sections, final figure) and the NetCDF of the zoom